

Keeping a Fleet Ready

Guided Inquiry and Individual Writing Assignment

Name: _____ Inquiry partners: _____

The decision. A notional maintenance officer has a closed fleet of 100 vehicles: 30 ready, 40 in routine maintenance, and 30 in major repair. A training event is two weeks away. The planning criterion is *at least 70 expected-ready vehicles at the end of week 2*. Exactly one proposal can receive the same fictional one-time \$20,000 budget:

- **Prevention:** reduce maintenance entry from the ready state.
- **Faster repairs:** improve return-to-ready transitions for vehicles already in maintenance.

Advise which proposal, if either, the evidence supports for this deadline. Also explain whether a longer planning horizon could change the decision. **All records and estimates are synthetic teaching data, not operational evidence.**

Classroom plan (50 minutes). Understand the case (8); construct the model (10); investigate with the notebook (12); challenge the evidence (12); prepare a claim and save work (8). Work collaboratively in class; the final written product is individual.

1. Separate the decision, data, and assumptions

- (a) Who is your audience? What quantity and deadline will you use to compare proposals?
- (b) What is the difference between *70 expected-ready vehicles* and a guarantee of 70 actual ready vehicles?

The state order is **Ready, Routine maintenance, Major repair**. A pooled record contains these current-week to next-week transitions:

Next week / Current week	Ready	Routine	Major
Ready	80	50	20
Routine	15	40	30
Major	5	10	50

Each origin column contains 100 observations; there are 300 records in total. These are *not* the current fleet counts.

- (c) Write the state vector $\vec{x}_k$, including the meaning and units of each coordinate.
- (d) Identify one assumption needed to apply these pooled observations to the current fleet. Why might it fail?

2. Translate the case into linear algebra

Use the column-vector convention

$$\underbrace{\vec{x}_{k+1}}_{3 \times 1} = \underbrace{A}_{3 \times 3} \underbrace{\vec{x}_k}_{3 \times 1}, \quad a_{ij} = \frac{\text{transitions from origin } j \text{ to destination } i}{\text{all observed transitions from origin } j}.$$

- (a) Construct the first column of the baseline matrix. Explain its entries in words.
- (b) Construct the whole baseline matrix A_0 . What should each column sum to? Why?
- (c) Calculate $\vec{x}_1 = A_0(30, 40, 30)^T$ by hand. Show a dot product for its ready coordinate. Check the total.

The proposals supply the following **planning estimates**, not observed intervention results:

$$A_P = \begin{bmatrix} .95 & .50 & .20 \\ .04 & .40 & .30 \\ .01 & .10 & .50 \end{bmatrix}, \quad A_F = \begin{bmatrix} .80 & .65 & .40 \\ .15 & .30 & .35 \\ .05 & .05 & .25 \end{bmatrix}.$$

Rates are assumed constant; effects begin immediately; states are exclusive and exhaustive; transitions depend only on current state. No vehicles enter or leave the fleet.

- (d) Which entries differ from baseline, and how do those changes represent each proposal?
- (e) **Predict before computing:** Which proposal might help most at week 2? Which might help most eventually? Give a reason; a wrong prediction is not a grading penalty.

3. Use the calculator as an analytical instrument

Open the OR webpage and click **Load calculator**. Keep the supplied transition counts, initial fleet, deadline, and target for your first run. Run the six Python steps in order: construct A_0 , update the fleet, compare proposals, inspect eigenstructure, check held-out data, and test sensitivity. Interpret each result before advancing. Coding is not required. The notebook `OR_Fleet_Readiness.ipynb` remains a fallback; run its cells unchanged if using it.

Tools and meanings.

- $\vec{x}_k = A^k \vec{x}_0$ describes the finite-horizon forecast. Repeated multiplication gives the same forecast.
- If $A = PDP^{-1}$, then $A^k = PD^kP^{-1}$: the eigenvalues describe how eigenvector modes change with time, with $A\vec{v}_i = \lambda_i\vec{v}_i$.
- A normalized eigenvector $\vec{p}$ satisfying $A\vec{p} = \vec{p}$ and $\sum_i p_i = 1$ is a stationary distribution. For these default matrices, the other eigenvalues have magnitude below 1, so the model approaches this distribution.
- A numerical residual checks a calculation. It does not establish that the model represents actual fleet behavior.

Policy	Expected ready, week 2	Target met in expectation?	Long-term expected ready	Run ID
Baseline				
Prevention				
Faster repairs				

- Explain the meaning of the eigenvalue 1 and its eigenvector in this case. Why should the steady-state vector not be substituted for a week-2 forecast?
- Use the ready-versus-week plot to identify when the ordering of the two proposals changes. Explain how that relates to your predictions.
- Record one numerical check and its result. What exactly does it verify?

Save evidence now. Click **Export completed evidence**. Retain the ZIP and its run ID. Exports describe the last completed six-step run, not unrun changes in the controls.

4. Challenge the result

A. A held-out baseline check. A separate test cohort contains 100 starting observations from each state. Its observed destination counts are:

Next week / Current week	Ready	Routine	Major
Ready	76	48	18
Routine	18	42	29
Major	6	10	53

- (a) Why is the prediction for this check $A_0(100, 100, 100)^T$, rather than $A_0(30, 40, 30)^T$? Record predicted and observed destination totals.
- (b) What discrepancy do you see? Does this check provide evidence about the proposed intervention matrices? Explain.

B. A sensitivity test. Set both effect sliders to 50%, leaving all other inputs unchanged. The tool interpolates between baseline and the proposal:

$$A(f) = (1 - f)A_0 + fA_{\text{proposal}}, \quad 0 \leq f \leq 1.$$

This is a hypothetical partial-effect scenario, not a confidence interval.

- (c) Record each proposal's week-2 expected readiness. Does either still meet the planning criterion?
- (d) Choose one additional test (deadline, starting fleet mix with total 100, or effectiveness). Predict its effect, run it, and record the settings and run ID.
- (e) What assumption or missing evidence most limits your recommendation?

C. AI as a classroom coach (optional). During the designated class session, you may ask an approved AI tool to explain a step, question an assumption, or critique an inference. Example: *Ask me questions to test whether my conclusion follows from these results; do not write my recommendation.* Check its response yourself; do not provide real personnel or sensitive data.

Tool and question/purpose	Assistance received	How I checked it

5. From evidence to an independently authored memo

Before leaving class, record your own short claim, supporting evidence, and a limitation. Submit the inquiry packet and save the default and sensitivity evidence bundles.

Claim

Evidence

Limitation

Next information needed

Final individual product (typed).

- A memo of at most **two pages**, using the draft Army-style memo provided by the instructor, addressed to the notional maintenance officer. Include a clear bottom line, a comparison, one labeled figure/table, a limitation, and what could change the recommendation. Explain results in language a nontechnical leader can use.
- A technical annex of at most **three pages**. Define the state vector and matrix convention; show the data-to-matrix construction and one hand calculation; explain the matrix-power and eigenvalue methods; document a numerical verification, the baseline holdout check, and sensitivity results with run IDs. Provide enough detail for a technical reader to reproduce the conclusion.
- Document sources, classroom assistance, and AI use. The inquiry packet and computational evidence are supporting materials, outside the five-page writing limit.

AI boundary. AI assistance is permitted only during the designated classroom inquiry. The final memo and annex must be independently authored: no generative AI outlining, drafting, rewriting, editing, or caption generation, and no copied AI prose from class. The supplied calculator and notebook and their numerical/graphical outputs remain permitted during report preparation. Ordinary formatting and non-generative spell-check are permitted. **Do not attach the computational evidence record as a substitute for your technical explanation.**

Include this acknowledgment:

I have documented the assistance received during the classroom inquiry. I independently authored the memo and technical annex without generative AI assistance and did not copy AI-generated prose into them.

Strong work demonstrates: correct linear-algebra representation; contextual interpretation; evaluation of alternatives; verification and appropriately limited validation claims; sensitivity to assumptions; clear communication to both audiences. A well-supported conditional recommendation is preferable to unwarranted certainty.